

Production II: Cost, and the Feedback

Advanced Microeconomics 1: Economic Decision Making. Lecture 6

Luke Heath Milsom

Today's route

1. The cost minimization problem — which you have already solved, in L2.
2. From $c(w, q)$ to supply: cost curves, $p = MC$, and the short run vs the long run.
3. Trace through the effects of an indexed wage rise.
4. What the cost function knows: c as a sufficient statistic for the technology.
5. What the competitive benchmark cannot see.

The cost minimization problem

The cost minimization problem

CMP

Given input prices $w \gg 0$ and required output q :

$$c(w, q) = \min_{z \geq 0} w \cdot z \quad \text{s.t.} \quad f(z) \geq q,$$

with solution $z(w, q)$, the **conditional factor demand**.

- “Conditional”: demand for inputs *conditional on producing* q . The scale decision is switched off; only the input-mix decision remains.
- Minimise expenditure subject to hitting a target level of an increasing function. This is the EMP with new labels.

The twin table

	Consumer (L2–L3)	Firm (today)
problem	$\min p \cdot x \text{ s.t. } u(x) \geq u$	$\min w \cdot z \text{ s.t. } f(z) \geq q$
value	$e(p, u)$: cost of living	$c(w, q)$: cost of producing
solution	$h(p, u)$: Hicksian demand	$z(w, q)$: conditional demand
Shephard	$\nabla_p e = h$	$\nabla_w c = z$
curvature	e concave in p	c concave in w
derivative matrix	$D_p h$ sym NSD, $D_p h p = 0$	$D_w z$ sym NSD, $D_w z w = 0$
law	compensated law of demand	compensated law of input demand

- The main difference: u was invisible and ordinal; q is **observable and cardinal**.
- $\Rightarrow$ today we will go fast. You've seen it all before!

Solving it: Lagrange, and what λ is now

- Interior FOC: $w_\ell = \lambda \partial f / \partial z_\ell$ for used inputs, so for any two:

$$\text{MRTS}_{\ell k} = \frac{w_\ell}{w_k}$$

the input-mix version of tangency. Same picture as L2, isoquant for indifference curve.

- And λ ? Same three steps as L2: $\lambda = \partial c(w, q) / \partial q$ — the cost of one more unit of output. λ is marginal cost.

Properties of c and z

Proposition · MWG 5.C.2 (selection)

Suppose $c(w, q)$ is the cost function of a continuous, increasing f , with $z(w, q)$ its conditional demand. Then:

- (i) c is **h.d.1** in w and nondecreasing in q ;
- (ii) c is **concave** in w ;
- (iii) z is **h.d.0** in w ; if f quasiconcave, $z(w, q)$ is convex-valued;
- (iv) **Shephard's lemma**: if $z(\bar{w}, q)$ is a singleton, $\nabla_w c(\bar{w}, q) = z(\bar{w}, q)$;
- (v) if z differentiable: $D_w z = D_w^2 c$ is **symmetric, NSD**, with $D_w z w = 0$.

Reading (ii) and (v): the 15% cut, as curvature

- Concavity of c in w says: when one input price rises, cost rises *less than linearly* — because the firm re-optimises the mix.
- As in the consumer section, with the signs flipped: the fixed basket over-compensates the household because it prices **no substitution**. Here: the fixed input mix *over-states* the cost of a shock because it prices **no substitution**.

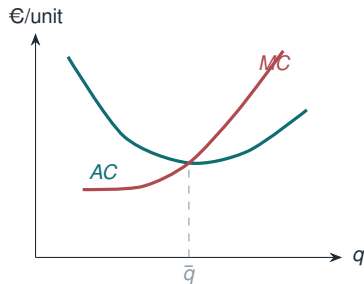
From cost to supply

From $c(w, q)$ to the cost curves

- Fix w ; write $C(q) = c(w, q)$. Define

$$AC(q) = \frac{C(q)}{q}, \quad MC(q) = C'(q).$$

- MC crosses AC at the minimum of AC : the **efficient scale** $\bar{q}$. (If AC is falling, the marginal unit is cheaper than the average, dragging the average down; and vice versa.)



NB: with CRS, $AC = MC = \text{constant}$ — the geometry degenerates.

Supply: $p = MC$, above AC

- Give the firm back its output choice, cost function in hand:

$$\max_{q \geq 0} p q - c(w, q) \quad \text{FOC: } p = MC(q)$$

- The competitive firm produces where **price equals marginal cost** — provided $p \geq AC$ at that output, otherwise it shuts down. The supply curve is the MC curve above AC .
- Anything that shifts MC shifts *supply*, and in the competitive benchmark, cost changes pass into prices through this equation.

What the cost function knows

c is a sufficient statistic for the technology

- Suppose you know $c(w, q)$ but not Y . Have you lost anything?
- Fix q . For each w , cost minimisation says the true input requirement set $V(q) = \{z : f(z) \geq q\}$ lies inside the half-space $\{z : w \cdot z \geq c(w, q)\}$. Intersect over all w :

$$V^*(q) = \{z \geq 0 : w \cdot z \geq c(w, q) \quad \forall w \gg 0\}.$$

- $V^*(q)$ is monotone, convex, and contains exactly the economically relevant part of $V(q)$: **the cost function encodes the technology**.
- We have seen this before; it is L4's recovery, but on the firm side. There: demand data $\rightarrow$ preferences. Here: the cost function $\rightarrow$ the production set.

Theorem · sufficient conditions

Let $\varphi(w, q)$ be differentiable, nondecreasing in w , h.d.1 in w , and concave in w . Then φ is the cost function of some well-behaved technology — namely the one with input requirement sets $V^*(q)$ built from φ .

- Compare L4: Slutsky symmetry + NSD + h.d.0 $\iff$ demand comes from *some* rational preference. Here: h.d.1 + concavity $\iff \varphi$ comes from *some* technology.

Sizing it: households, with the firm switched on

The setup

- Belgian wages are indexed. When the 2022 shock raised consumer prices, wages rose automatically.
- But the same event, from the firm's perspective: an economy-wide rise in w_L .
- Unequal incidence: Wages are indexed by CPI; firm costs rose by less due to import-inflation.

Five channels, five signs

1. **Cost, first order:** $\frac{\partial c}{\partial w_L} = z_L > 0$ (Shephard). Costs rise by exactly the labour bill.
2. **Substitution:** $\frac{\partial z_L}{\partial w_L} \leq 0$ (concavity). Labour per unit of output falls — capital, energy, outsourcing.
3. **Scale:** $MC \uparrow$ at every q , so $p = MC$ moves the optimum: $q^* \downarrow$. Labour falls *again*, through output.
4. **Price:** in the competitive benchmark the industry supply shifts up, and the new equilibrium p is *higher*. Part of the wage rise returns to households as **inflation**.

From signs to sizes

- The five channels are signed. We want **magnitudes**. Use a small calibrated model of the average firm.
- Assumptions: **A1** CES cost function (the form the literature uses for this shock: Bachmann et al. 2022); **A2** constant returns; **A3** price taking; **A4** industry demand elasticity $\varepsilon_D = -1$.

$s_L = 0.11$	labour share of variable cost	108k Belgian firms	Bijmens et al.
$s_O = 0.89$	all other inputs	$= 1 - s_L$	—
$\sigma = 0.4$	substitution elasticity	from the employment evidence	BHMN back-out
$g_O = 5.9\%$	other input costs, 2022		—
g_O includes producer energy +93% (ECB) at its $\approx 3\%$ cost weight (EC energy report).			
$g_w = 9.6\%$	cheque = 2022 avg CPI		Statbel

Channel 1

A1, written down, two inputs, labour L and everything else O :

$$c(w_L, w_O, q) = q \Phi(w)^{\frac{1}{1-\sigma}}, \quad \Phi(w) \equiv a_L w_L^{1-\sigma} + a_O w_O^{1-\sigma}.$$

Take logs, differentiate with respect to $\ln w_i$:

$$\frac{\partial \ln c}{\partial \ln w_i} = \frac{1}{1-\sigma} \cdot \frac{(1-\sigma) a_i w_i^{1-\sigma}}{\Phi} = \frac{a_i w_i^{1-\sigma}}{\Phi} \stackrel{\text{Shephard}}{=} \frac{w_i z_i}{c} = s_i.$$

So, to a first-order:

$$d \ln c = s_O g_O + s_L g_w = 5.2\% + 1.1\%$$

Channel 1: costs rise by exposure $\times$ shock, before any behaviour.

Channel 1. Uses: Shephard. Assumes: A1.

Channel 4, the price

Under **A2** (CRS), cost is proportional to output: $c = q \varphi(w)$ with $\varphi(w) = c(w, 1)$. Hence

$$AC = MC = \varphi(w), \quad \text{flat in } q.$$

Under **A3**, an interior optimum requires $p = MC$; with $AC = MC$ this is also *zero profit*. This **pins the price** to input costs,

$$p_O = \varphi(w) \implies \boxed{d \ln p_O = d \ln c = 5.2\% + 1.1\%}$$

Channel 4: the cost rise passes into the output price one-for-one — and the 1.1pp piece is the policy returning to households as inflation.

Channel 4. Uses: $p = MC$; CRS. Assumes: A2, A3.

Channel 2

Shephard, computed on A1: $z_L = \partial c / \partial w_L = q a_L w_L^{-\sigma} \Phi^{\frac{\sigma}{1-\sigma}}$. Log-differentiate, using $\partial \ln \Phi / \partial \ln w_j = (1 - \sigma) s_j$:

$$\frac{\partial \ln z_L}{\partial \ln w_L} = -\sigma(1 - s_L), \quad \frac{\partial \ln z_L}{\partial \ln w_O} = +\sigma s_O = +\sigma(1 - s_L).$$

Equal and opposite — so only the *difference* of the shocks matters:

$$d \ln z_L|_q = -\sigma(1 - s_L)(g_w - g_O) = -0.4 \times 0.89 \times 3.7\text{pp} = -1.3\%$$

- **Why does the firm adjust labour at all?** If wages and other inputs rose *uniformly*, $g_w = g_O$, the box says **zero**: z is h.d.0 in w — the firm-side no-money-illusion theorem.
- The mix moves only because indexation made labour *relatively* dear: the CPI contains imported prices firms never receive, so $g_w = 9.6$ vs $g_O = 5.9$ — a **3.7pp wedge**.

Channel 2. Uses: Shephard; h.d.0. Assumes: A1.

Channel 3

Unconditional labour is $L = z_L(w, q^*)$; under A2, $z_L \propto q$, so

$$d \ln L = d \ln z_L|_q + d \ln q^*, \quad \text{and } \mathbf{A4} \text{ closes output: } d \ln q^* = \varepsilon_D d \ln p_O.$$

$$d \ln L = \underbrace{-\sigma(1 - s_L)(g_w - g_O)}_{\text{mix: } -1.3\%} + \underbrace{\varepsilon_D(s_O g_O + s_L g_w)}_{\text{scale: } -6.3\%} = -7.6\%$$

- Channel 3: more expensive output $\Rightarrow$ less sold $\Rightarrow$ less of *every* input, labour twice over.

Channel 3. Uses: CRS; A4. Assumes: partial equilibrium — the GE frame brackets it.

The firms reaction

	2022 cost shock	shock + indexation	the policy's addition
other-goods prices p_o	+5.2%	+6.3%	+ 1.1%
employment	-3.1%	-7.6%	- 4.5%

- Wage indexation exacerbated things from the firms' side.
- Perhaps unexpectedly the main incidence lies on employment.

Before going back to households: one inconsistency to fix

- Block 1 priced only *energy*; but the cheque is indexed to *everything*.
- The fix: let the other price move too. Define π_O so the *average* basket reproduces observed 2022 inflation:

$$0.064 \times 57.9\% + 0.936 \pi_O = 9.6\% \implies \pi_O = 6.3\%.$$

- And the model just told us where π_O comes from:

$$\underbrace{6.3\%}_{\pi_O} = \underbrace{5.2\%}_{\substack{\text{other input costs through firms, } s_{OGO} \\ \text{(incl. producer energy +93\% at } \approx 3\% \text{ weight)}}} + \underbrace{1.1\%}_{\substack{\text{the policy itself} \\ s_{LGW}}}$$

- So block 2 has **opened block 1's black box**: part of “other-goods inflation” was the energy shock arriving through firms — and part was the repair arriving as disease. Without the policy, π_O would have been 5.2%.

The household's bill, at full prices

Stone–Geary again, both prices moving ($\tilde{\gamma} \equiv p_E^0 \gamma / w$, the subsistence share):

$$\frac{-CV}{w} = \tilde{\gamma} g_E + (1 - \tilde{\gamma}) \left[(1 + g_E)^\beta (1 + \pi_O)^{1-\beta} - 1 \right]$$

	full bill	cheque (= avg CPI)	net
poor, $s_0 = 0.10$	11.3%	9.6%	−1.7%
rich, $s_0 = 0.06$	9.3%	9.6%	+0.3%

L4's one-number lesson, reproduced at full prices: the single rate under-compensates the exposed and over-compensates the sheltered — same sign, same size (± 1 – 2 pp).

Proof → p. 32

The bottom line

	no indexation	automatic indexation
firms exogenous (block 1)	(a) real wage: poor −11.3% , rich −9.3% (b) employment: unchanged	(a) real wage: poor −1.7% , rich +0.3% (b) employment: unchanged
firms endogenised (this block)	(a) real wage: poor −10.4% , rich −8.3% (prices at the 5.2% counterfactual) (b) employment: −3.1%	(a) real wage: poor −1.7% , rich +0.3% (b) employment: −7.6% — −4.5 from policy

Across the top: exogenous firms see the cheque as free — real wages repaired, nothing else moves. Down the columns: endogenising firms adds the two things block 1 could not see — the **jobs** column, and the fact that part of each bill **is the policy** (+1.1pp of π_O).

The bottom line

- Endogenising firms reveals the real losers from both inflation and indexation → those who became unemployed.
- The policy does a fairly good job of sheltering the remaining employed.
- Two things we haven't (and won't) discuss:
 1. Indexation has to be paid for — taxation.
 2. Those extra wages are spent — GE loop back to demand.

Why indexation fails, on both sides of the market

- **On the uniform component, it does not fail.** If every price and the wage rose by the same π , households are exactly compensated (h h.d.0 — L1) and firms change nothing (z h.d.0 — today). **A perfect hedge against uniform inflation.**
- **Consumer side (L4): a measurement failure.** The cheque prices no substitution, and one rate meets many baskets: ± 1 –2pp across households.
- **Firm side (today): an incidence failure.** The CPI contains imported prices firms never receive, so indexation raises labour's *relative* price — the 3.7pp wedge — and so unemployment rises.

A reality check

- The model implies employment fell 7.6%! Did it? No — it actually rose.
- What went wrong with our modeling?
 1. Other things happened — post-COVID recovery.
 2. GE — indexation boosts demand as the money is spent, we need to endogenise demand.
- Real-life is (even more) complex!

What we have yet to consider

- But there's more!
- Thus far we have only considered things from a static perspective.
- The cost of inflation is not just price rises today, but the **threat of future rises**.
- Wage indexation insures (the employed) against these rises. **What is the insurance value** of the policy?
- To answer this we need to consider, choice under uncertainty.

Appendix

Derivation: the two-price Stone–Geary bill

Stone–Geary, $u = \beta \ln(x_E - \gamma) + (1 - \beta) \ln x_O$. Its expenditure function (solve the EMP; or integrate Shephard twice):

$$e(p_E, p_O, u) = p_E \gamma + \kappa p_E^\beta p_O^{1-\beta} e^u, \quad \kappa = \beta^{-\beta} (1 - \beta)^{-(1-\beta)}.$$

Evaluate at u^0 using $e(p^0, u^0) = w$: the supernumerary part is worth $w - p_E^0 \gamma$ at old prices, and it scales by the price ratios:

$$e(p^1, u^0) = p_E^1 \gamma + (w - p_E^0 \gamma) \left(\frac{p_E^1}{p_E^0} \right)^\beta \left(\frac{p_O^1}{p_O^0} \right)^{1-\beta}$$

Subtract w , divide by w , write $\tilde{\gamma} = p_E^0 \gamma / w$:

$$\frac{-CV}{w} = \tilde{\gamma} g_E + (1 - \tilde{\gamma}) \left[(1 + g_E)^\beta (1 + \pi_O)^{1-\beta} - 1 \right].$$

Set $\pi_O = 0$: L4's one-price formula, exactly. The subsistence quantity γ is priced at the *full* energy rise; only the flexible remainder substitutes.